
Where a Reconstructed Magnetized-CBET Vlasov Model Breaks: A Preregistered Vlasov–PIC Boundary Map

Anonymous Authors

Abstract

Magnetized cross-beam energy transfer (MagCBET) is inexpensive to evaluate with linear kinetic theory and expensive to validate with particle-in-cell (PIC) simulation, but a usable boundary between those descriptions has not been reported. We preregister one agreement score, one 30% threshold, three mechanism tests, and a registered 20-cell map spanning magnetic field, field orientation, pump intensity, and polarization. Eighteen cells reanalyze the published EPOCH archive; two intensity endpoints are new 2D–3V EPOCH conditions, backed by a magnetic-field trend, deterministic duplicate, and resolution run. A separately frozen pre-outcome refinement adds one near-reference intensity condition, giving 21 reported cells. The baseline is an explicitly partial, uncalibrated reconstruction of the 2026 MagCBET Vlasov model, with magnetized Maxwellian susceptibilities, cold laser eigenmodes, coherent mode propagation, and adjacent finite-box modes. Under the frozen interval rule, 0 cells hold, 12 break, 6 are fuzzy, and 3 are comparison-invalid; 18/21 cells have established support. The reduced suite passed every reduced-scale gate, and the new intensity spoke does not supply an established hold/break transition. Only 5 breaks satisfy a preregistered physical-mechanism conjunction; the others remain explicitly unattributed or are diagnosed as comparison failures. All three new conditions exceed the 0.05 parasitic-light limit (range 0.189–0.268), so none establishes a new theory boundary. The compactness audit is non-identifiable because the usable cells contain 0 holds and 12 breaks. These results give a decision map for the documented reconstruction, while sharply delimiting what cannot yet be inferred about the inaccessible companion implementation.

1 Introduction

Cross-beam energy transfer (CBET) redistributes laser power through an ion-acoustic beat wave. It is deliberately used to tune indirect-drive symmetry [Michel et al., 2009, Moody et al., 2012], yet it can lower direct-drive coupling, motivating wavelength-detuning mitigation [Marozas et al., 2018]. This is no longer an unmagnetized design problem: megagauss fields occur in magnetized inertial-confinement-fusion (ICF) platforms [Moody et al., 2022], and a recent hohlraum experiment directly connected polarization-dependent CBET to asymmetric backscatter [Hao et al., 2025].

Shi and Moody’s PIC study established the qualitative MagCBET signatures: parallel-polarized gain falls with field, orthogonal transfer becomes allowed, the laser eigenmodes split, and near-perpendicular ion response shifts in detuning [Shi and Moody, 2026]. Their companion Vlasov theory provides a fast prediction [Moody and Shi, 2026], but the PIC comparison retained quantitative differences. A qualitative trend check does not answer the engineering question: *where may a Vlasov estimate be used, and where is PIC still required?*

The distinction matters because a theory/PIC residual has two meanings. It may reveal a physical approximation—the warm-fluid drive, the dropped nonlinear ion force, or the single strongly damped ion-acoustic-wave (IAW) picture—or it may only reveal an invalid comparison caused by drift, depletion, scattering, noise, or an under-resolved box. CBET is known to saturate through ion heating and pump depletion [Hansen et al., 2021, 2022]; nonlinear kinetic calculations find trapping, spectral shifts, and harmonics [Seaton et al., 2022, Yin et al., 2023]. A 2026 sub-nanosecond study likewise separates moderate-intensity depletion from a strongly coupled regime with IAW harmonics and trapping [Xu et al., 2026]. We therefore preregister comparison-validity tests *before* physical attribution.

Our contribution is a cellwise boundary map, not a new closure. It differs from earlier careful warm-fluid/PIC benchmarking [Shi, 2023] by freezing a deployment-oriented error rule, mapping multiple axes, propagating cell uncertainty into the classification, and allowing the honest outcomes “fuzzy,” “comparison-invalid,” and “mechanism-unattributed.” Every number below is generated from a corresponding JSON/CSV or raw-SDF record.

2 Preregistered question and design

The frozen files `proposal/code/sweep_plan.{json,md,py}` carry the declaration time 14:21:45 UTC on 31 July 2026 and were hash-frozen before the archive loader was run at 14:31:24 UTC and before EPOCH was launched. The carbon reference is $n_e = 3.5 \times 10^{20} \text{ cm}^{-3}$, $T_e = 0.8 \text{ keV}$, $T_i = 0.2 \text{ keV}$, $\lambda_0 = 0.351 \mu\text{m}$, a 30° beam crossing, and positive $0.9 \text{ \AA}$ seed detuning. The core is the full product

$$|B| \in \{2, 10, 50\} \text{ MG}, \quad \phi_B \in \{-13, 30, 75\}^\circ, \quad p \in \{zz, zy\}, \quad (1)$$

at pump and seed intensities $0.5 \times 10^{15} \text{ W cm}^{-2}$. The field azimuths correspond to beat-wave/field angles of approximately 88° , 45° , and 0° . A targeted zz spoke at $(10 \text{ MG}, 30^\circ)$ uses pump intensities $\{0.25, 0.50, 1.00\} \times 10^{15} \text{ W cm}^{-2}$ with fixed seed intensity; its two endpoints are new. Thus B , angle, and intensity each have three levels. Polarization is a preregistered categorical contrast and is not entered as a continuous coordinate in the compactness model.

After dump 2 but before either late-time dump existed in any new run, we froze a one-cell refinement at $0.60 \times 10^{15} \text{ W cm}^{-2}$ with otherwise identical center parameters. The published study had already warned that the low endpoint needs a longer interaction length and that the high endpoint admits noise-seeded competition [Shi and Moody, 2026]; the refinement was therefore a comparison-validity anchor, not an outcome-selected boundary point. Its timestamp, rationale, parameters, and hashes are in `refinement_plan_001.{json,md}`. We include it regardless of outcome, mark it separately from the registered 20 cells, and leave the score, threshold, mechanism tests, and registered three-point intensity conjunction unchanged.

2.1 One comparison rule

The logged energy gain $g = \log(U/U_0)$ is mapped to exponential-envelope gain G by the finite-length identity

$$g = \log\left(\frac{e^G - 1}{G}\right), \quad \kappa = \frac{G}{L_x/\lambda_0}. \quad (2)$$

The sole score is

$$A = \frac{|\kappa_V - \kappa_{\text{PIC}}|}{\max(|\kappa_{\text{PIC}}|, 4 \times 10^{-4})}. \quad (3)$$

The floor prevents a nearly forbidden channel from dominating by division by zero. We call a cell *holds* only when the largest A over the full PIC uncertainty interval is at most 0.30, *breaks* only when the smallest is above 0.30, and *fuzzy* otherwise. There was no search over scores or thresholds. Reported spreads are conservative ranges, not standard errors, so we do not attach asymptotic p -values.

3 Methods

3.1 Documented partial Vlasov reconstruction

The published companion’s bibliographic record was available, but its body was not accessible through the publisher routes available in this environment. We therefore adopt a partial reconstruction and never claim bitwise or equation-level identity with that implementation. The baseline combines standard Maxwellian kinetic response [Fried and Conte, 1961, Stix, 1992] with the published magnetized three-wave framework [Shi et al., 2017, Shi, 2019]. With $v_{Ts} = \sqrt{2T_s/m_s}$, $\zeta_{ns} = (\omega - n\Omega_s)/(k_{\parallel}v_{Ts})$, and $b_s = k_{\perp}^2 T_s/(m_s \Omega_s^2)$, we evaluate

$$\chi_s = \frac{1}{k^2 \lambda_{Ds}^2} \left[1 + \frac{\omega}{k_{\parallel} v_{Ts}} \sum_{n=-N}^N e^{-b_s} I_n(b_s) Z(\zeta_{ns}) \right]. \quad (4)$$

We use the fixed b_s -dependent truncation $N = \max[12, \lceil b_s + 9\sqrt{\max(b_s, 1)} + 24 \rceil]$; requests above 12,000 harmonics use the analytic unmagnetized continuum limit. The retained driven density response is

$$\mathcal{R} = k^2 \lambda_{De}^2 \frac{\chi_e(1 + \chi_i)}{1 + \chi_e + \chi_i}. \quad (5)$$

We solve $1 + \chi_e + \chi_i = 0$ from three complex IAW-like initializations for each of the nine (B, ϕ_B) settings and log every attempt and residual.

For each laser, a cold-plasma dielectric eigenproblem supplies complex electromagnetic polarizations. A linear launch is propagated as a *coherent* sum of the two modes before evaluating the electrostatic-like warm-fluid ponderomotive overlap at the averaged laser frequency. The SVEA prefactor is evaluated from SI field amplitudes $I = \epsilon_0 c |E|^2/2$; no PIC value fixes its scale. The drive is linear in pump intensity and omits ion ponderomotive nonlinearity and depletion, as the tested model requires. Finally, the incident seed is partitioned between two adjacent cold-linear longitudinal box modes separated by $2\pi/L_x$ and their energies are averaged. This is a reduced form of the supplied finite-box routine: without the companion pump-on electromagnetic dispersion function, it omits the nonlinear real-wave-number shift and associated interference bounds. Source, roots, harmonic terms, eigenvectors, overlaps, adjacent modes, and 1278 curve evaluations are retained in the logs.

3.2 Published PIC archive

The archive [Shi, 2025] supplies the original $500\lambda_0 \times 10\lambda_0$, 20-cell/ λ_0 , 20-particle/cell carbon simulations. We read all registered rows of `transfer_{zz,zy}_dL0.9.txt` without deletion or outcome recursion. Those files explicitly use the supplied `Gain_Fourier_Integral.m`: dumps 1, 5, and 6, both seed peaks, $\Delta k_x = 0.03|k_x|$, and $|k_y| \leq 3\Delta k_y$.

The archive contains processed gains rather than the raw fields needed to recompute our drift, pump-strip, parasitic-spectrum, and SNR prerequisites. For those cells, “comparison valid” therefore inherits the source study’s documented linear-regime selection; physical attributions from archive cells are conditional on that provenance claim. New runs, by contrast, are gated from their retained raw fields.

The raw estimator contains propagation/scatter loss. Before cell comparison, one common correction was estimated from all eight detunings of the forbidden $B = 0$, zy channel: $g_0 = -0.2342605$, detuning standard deviation 0.0061814, reported-spread RMS 0.0800576, and quadrature uncertainty 0.0802959. Raw and corrected columns and every input hash remain available. Treating the reported full spread as a symmetric interval is deliberately more conservative than treating it as a half-range.

3.3 New reduced EPOCH suite

We compiled the supplied EPOCH 2D–3V source at commit `f294c484f76d` (the full source hash is retained in the build log) [Arber et al., 2015]. The registered box is $160\lambda_0 \times 10\lambda_0$, with 12 cells/ λ_0 , 12 particles per cell, 0.95 CFL, collisionless fully ionized carbon, and the original simple-laser/open

CPML arrangement. The seed enters from $x_{\min}$; the pump enters from $x_{\min}$ and $y_{\max}$ at 30° and starts at 0.6 ps. Six field/density snapshots span 0.6–3.6 ps. A 16-cell/ λ_0 , 16-particle/cell center run tests convergence. All cases use two MPI ranks.

The source fixes the RNG to 7842432 + rank when `use_random_seed=false`. A complete duplicate tests that determinism; failure at 10^{-10} in processed gain would have triggered three clock-seeded repetitions at every new point. After registration but before any late-time output was available, we added a pump-off center control to give a matched background sample for every late-time/transverse-fraction pair. It is a disclosed diagnostic, not an added sweep cell or a changed threshold. The separately frozen 0.60-intensity refinement described above uses the same deck template, diagnostics, and unconditional reporting policy as the two registered endpoints.

Our Python port reads native SDF through the official SDF C library and reproduces the total Fourier integral. It evaluates central transverse fractions $f_y = 1, 0.75, 0.5$ at 3.0 and 3.6 ps relative to 0.6 ps. The six control-subtracted gains yield their mean and full max-to-min spread. We do not replace this registered total-energy metric with a more favorable eigenmode metric. As a parity diagnostic, we additionally port the supplied real-space mode path: a paired Gaussian seed filter, cold-plasma transverse eigenvectors, the Faraday shift $E_z(x - h)$ with $h = \psi/\bar{k}$, and separation of both real mode fields. The authors’ driver calls a warm-polarization package that is not supplied, so our logged branch values use the reconstructed cold eigenvectors and are not claimed as mode-resolved boundary measurements.

Four validation gates were frozen: the reduced zz coefficient at $\phi_B = 30^\circ$ must decrease from 2 to 10 to 50 MG; its 10-MG value must be within 35% of the archive; seed-window SNR at 3.0 ps must be at least five; and the 16/16 run must change the center coefficient by at most 20%. Failure makes the intensity endpoints illustrative, not a test of Vlasov physics. Separately, the deterministic duplicate must agree within 10^{-10} ; failure would invoke the frozen clock-seeded replication fallback rather than be silently treated as independent evidence.

3.4 Mechanism and comparison diagnostics

Comparison validity has precedence. We flag late drift above 25%, filtered pump depletion/side loss above 10%, non-pump/non-seed optical spectral energy above 5%, SNR below five, or a failed validation gate. Spectral and spatial windows were fixed in the analysis source: six equal-area SNR windows at k_y offsets $\pm 12, \pm 24, \pm 36$ bins, pump half-width $0.04k_0$, and upstream and downstream x/L_x strips $[0.05, 0.25]$ and $[0.75, 0.95]$.

The frozen plan did not specify a numeric sheath-field gate. Before any late snapshot existed, we added a non-decision effect size: the combined RMS of E_x in the outer 2% of the x faces and E_y in the outer 2% of the y faces, normalized by the vacuum seed field. These boundary-normal components exclude the injected z polarization but can still contain mode-rotated light; the ratio is therefore a sheath-sensitive proxy, not a unique sheath measurement. It is reported without changing a class or mechanism label.

Three physical conjunctions were preregistered. (H1) The warm-fluid drive is assigned only when $|\Delta\omega|/\omega_{ce} \geq 0.1$ or $\omega_{ce}/\bar{\omega} \geq 0.1$, the near-parallel cell breaks, and no new narrow peak appears. (H2) The single-IAW/unperturbed-orbit approximation is assigned only above 80° when an extra peak or a normalized curve centroid/width residual exceeds 0.30. Curve moments use positive weight above each curve’s minimum; peaks require 10% prominence. (H3) Dropped nonlinear ion response is assigned only when A rises monotonically along intensity and $P(2\Delta k)/P(\Delta k)$ crosses from below 0.10 at the center to above 0.10 at high intensity. A break satisfying none is “mechanism-unattributed.” These deterministic tests were frozen as a family; no post-hoc significance test or multiplicity-sensitive selection is used.

3.5 Compactness

For non-fuzzy established cells, we attempt to fit all seven nonempty subsets of $\log_{10} B$, $\angle(\Delta\mathbf{k}, \mathbf{B})$, and $\log_{10} I_p$ with binomial logistic regression. Models are compared by leave-one-out balanced accuracy. The fewest-axis model within 0.05 of the full model is selected; one or two axes constitutes a compact boundary. Spearman correlations with continuous A are descriptive effect sizes. When fewer than four cells or only one class survives the support filter, the classifier is reported as

Boundary map; cell text is registered score A

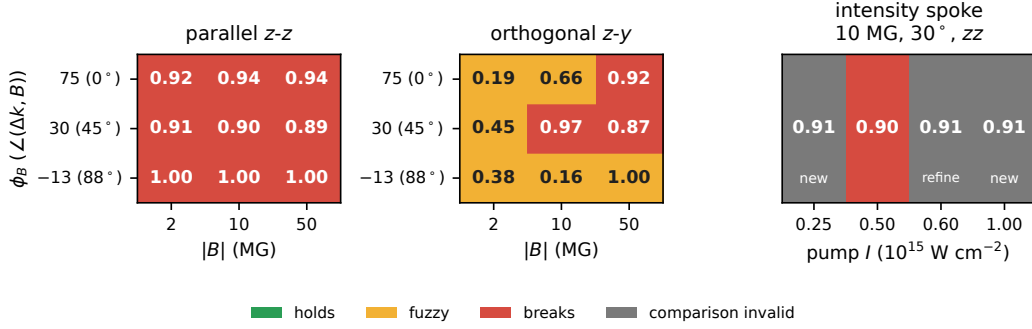


Figure 1: The registered boundary map plus pre-outcome refinement. Colors apply the full uncertainty- interval rule; text is the center score A . The core panels project onto the field-strength/orientation plane while retaining polarization. “New” marks registered reduced-EPOCH endpoints and “refine” marks the separately frozen pre-outcome condition on the intensity spoke.

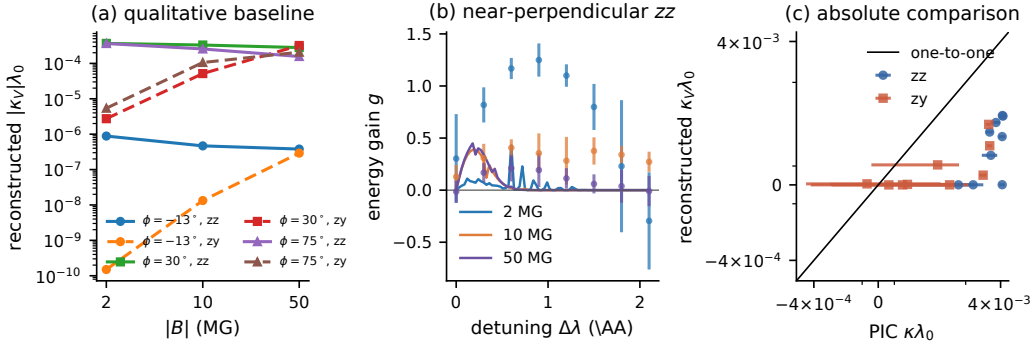


Figure 2: Baseline and archive. (a) The partial reconstruction reproduces decreasing zz gain and magnetically enabled zy gain. (b) At near- perpendicular geometry, points are corrected archive gains with conservative uncertainties and lines are uncalibrated theory. (c) Absolute coefficient comparison; axes are symmetric-logarithmic inside the registered floor.

non-identifiable and no one-class fits are manufactured. This was specified before the labels were available.

4 Results: the boundary map

Figure 1 and Table 1 contain the complete test set. Of 21 cells, 0 hold, 12 break, 6 are uncertainty-limited, and 3 fail comparison prerequisites. We label 18 established and 3 illustrative. This distribution is itself the boundary: the floor preserves physically suppressed channels without allowing their relative residual to diverge, while energetic channels demand quantitative agreement.

Support, rather than a hidden axis choice, controls the amber region: 6/6 fuzzy archive cells are in the orthogonal-polarization channel, whose uncertainty intervals reach the registered floor or cross zero, whereas all 9 parallel-polarization archive cells robustly break. Thus the archive plane contains no demonstrated absolute-agreement island for the reconstruction; the weak channels are uncertainty-limited rather than evidence of agreement.

The qualitative checks in Fig. 2a all pass for the stated signatures: zz decreases at every angle, zy increases at 75° , and the low-field near-perpendicular zy channel remains negligible. The near-perpendicular resonance centroid moves from $0.504 \text{ \AA}$ at 2 MG to $0.248 \text{ \AA}$ at 50 MG. Its reconstructed RMS width, however, decreases from 0.295 to $0.131 \text{ \AA}$, opposite the published broadening

trend. That failed qualitative check is important: the absolute map tests the documented partial baseline, not a surrogate asserted to be the full companion code.

The absolute discrepancy is not localized to one magnetic geometry. Among the 13 archive cells above the registered coefficient floor, the reconstructed-to-PIC coefficient ratio has median 0.060 and range 0.000208–0.128. This broad low bias makes model provenance, the reduced finite-box step, and omitted drive terms competing explanations for every local physical conjunction; we therefore report the frozen labels but do not reinterpret them as validation of the inaccessible companion implementation.

Table 1: Complete boundary evaluation: the registered design plus the labeled pre-outcome refinement. Prefixes P, N, and R denote archive, registered-new, and refinement cells; ‘invalid’ means comparison-invalid. Parameter tuples are $(|B| \text{ [MG]}, \phi_B [^\circ], n_e [10^{20} \text{ cm}^{-3}], T_e \text{ [keV]}, I_p [10^{15} \text{ W cm}^{-2}], \text{pol.})$. g is energy gain; PIC uncertainties are conservative full spreads. The score column gives its uncertainty interval.

Cell	$ B , \phi_B, n_e$ $T_e, I_p, \text{pol.}$	g_V	g_{PIC}		A class	mechanism
P01	2, −13, 3.5	2.21×10^{-4}	1.249 ± 0.160	0.9998 [0.9998, 0.9998]	breaks	single-IAW/orbit
P02	0.8, 0.5, zz 10, −13, 3.5	1.17×10^{-4}	0.355 ± 0.190	0.9997 [0.9993, 0.9998]	breaks	single-IAW/orbit
P03	0.8, 0.5, zz 50, −13, 3.5	9.50×10^{-5}	0.193 ± 0.163	0.9995 [0.3004, 0.9997]	breaks	single-IAW/orbit
P04	0.8, 0.5, zz 2, −13, 3.5	3.75×10^{-8}	0.038 ± 0.109	0.3773 [0.0000, 1.0000]	fuzzy	not applicable
P05	0.8, 0.5, zy 10, −13, 3.5	3.33×10^{-6}	0.017 ± 0.127	0.1647 [0.0000, 1.0000]	fuzzy	not applicable
P06	0.8, 0.5, zy 50, −13, 3.5	7.30×10^{-5}	0.135 ± 0.156	0.9994 [0.0000, 0.9997]	fuzzy	not applicable
P07	0.8, 0.5, zy 2, +30, 3.5	0.093	1.260 ± 0.164	0.9145 [0.9033, 0.9231]	breaks	unattributed
P08	0.8, 0.5, zz 10, +30, 3.5	0.084	0.945 ± 0.191	0.9007 [0.8782, 0.9157]	breaks	unattributed
P09	0.8, 0.5, zz 50, +30, 3.5	0.071	0.716 ± 0.105	0.8919 [0.8750, 0.9045]	breaks	unattributed
P10	0.8, 0.5, zz 2, +30, 3.5	6.85×10^{-4}	0.046 ± 0.120	0.4467 [0.0000, 0.9957]	fuzzy	not applicable
P11	0.8, 0.5, zy 10, +30, 3.5	0.013	0.541 ± 0.085	0.9742 [0.9697, 0.9775]	breaks	unattributed
P12	0.8, 0.5, zy 50, +30, 3.5	0.081	0.689 ± 0.081	0.8716 [0.8559, 0.8840]	breaks	unattributed
P13	0.8, 0.5, zy 2, +75, 3.5	0.093	1.293 ± 0.153	0.9171 [0.9073, 0.9248]	breaks	unattributed
P14	0.8, 0.5, zz 10, +75, 3.5	0.065	1.247 ± 0.178	0.9399 [0.9311, 0.9465]	breaks	unattributed
P15	0.8, 0.5, zz 50, +75, 3.5	0.039	0.755 ± 0.244	0.9423 [0.9174, 0.9552]	breaks	drive window
P16	0.8, 0.5, zz 2, +75, 3.5	1.35×10^{-3}	-0.017 ± 0.116	0.1876 [0.0000, 1.0135]	fuzzy	not applicable
P17	0.8, 0.5, zy 10, +75, 3.5	0.027	0.094 ± 0.104	0.6576 [0.0000, 0.8619]	fuzzy	not applicable
P18	0.8, 0.5, zy 50, +75, 3.5	0.052	0.719 ± 0.084	0.9202 [0.9106, 0.9279]	breaks	drive window
N19	0.8, 0.5, zy 10, +30, 3.5	0.016	0.188 ± 0.040	0.9113 [0.8878, 0.9265]	invalid	prereq. failure
R20	0.8, 0.25, zz 10, +30, 3.5	0.039	0.473 ± 0.110	0.9117 [0.8866, 0.9273]	invalid	prereq. failure
N21	0.8, 0.6, zz 10, +30, 3.5	0.066	0.778 ± 0.157	0.9071 [0.8858, 0.9214]	invalid	prereq. failure
Compactness summary		not identifiable; LOOCV balanced accuracy not defined under the frozen within-0.05 rule.				

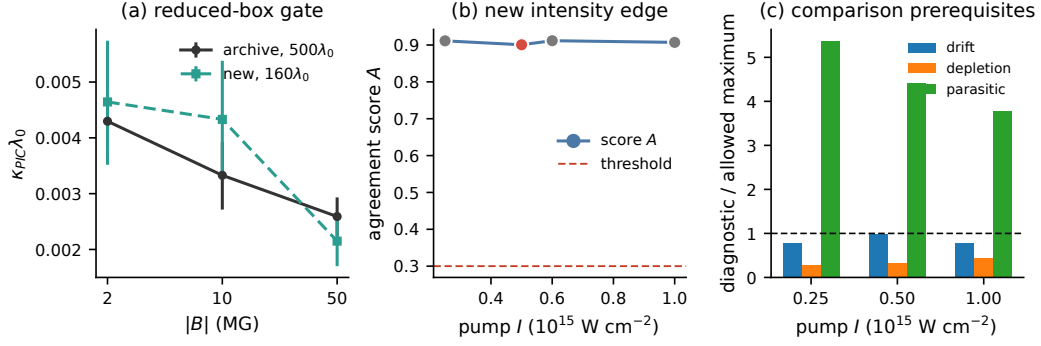


Figure 3: New EPOCH evidence. (a) Reduced and archive magnetic-field trends. (b) Frozen score along the registered spoke plus the labeled pre-outcome refinement; color is the final support-aware class. (c) Comparison diagnostics normalized by their preregistered maximum; values above one fail.

Table 2: Reduced-EPOCH measurements and comparison-prerequisite diagnostics. Resolution is cells/ λ_0 /particles per cell; uncertainties are full six-realization spreads.

run	MG	res.	g	$\kappa\lambda_0$	drift	SNR	parasitic	depletion
validation_B2	2	12/12	0.394 ± 0.100	0.004644	0.243	3195	0.219	0.036
validation_B10	10	12/12	0.366 ± 0.095	0.004331	0.245	3076	0.221	0.032
validation_B50	50	12/12	0.177 ± 0.038	0.00215	0.178	2116	0.232	0.000
intensity_I0p25	10	12/12	0.188 ± 0.040	0.002284	0.192	2488	0.268	0.029
intensity_I0p60	10	12/12	0.473 ± 0.110	0.005506	0.224	3169	0.211	0.037
intensity_I1p0	10	12/12	0.778 ± 0.157	0.00873	0.197	4670	0.189	0.044
resolution_B10	10	16/16	0.381 ± 0.115	0.0045	0.255	6492	0.199	0.000
duplicate_B10	10	12/12	0.366 ± 0.095	0.004331	0.245	3076	0.221	0.032

4.1 Reduced-PIC gate, new intensity cells, and refinement

The reduced suite passed every reduced-scale gate. Its center differs from the archive by 30.1%, and the 16/16 check changes the coefficient by 3.9%. The duplicate processed-gain difference is 0. The measured reduced coefficients at 2, 10, and 50 MG are 0.00464, 0.00433, and 0.00215 per λ_0 , respectively. These facts, rather than visual similarity, enact the frozen gate.

Passing the reduced-scale gate does not make every reduced run a valid theory comparison. The 16/16 diagnostic itself has late drift 0.255, just above the 0.25 cellwise limit, while its frozen resolution decision is based on the 3.9% coefficient change. More decisively, all three genuinely new conditions fail the independently frozen parasitic-spectrum prerequisite.

At pump intensities 0.25 and 1.00 in units of 10^{15} W cm⁻², the new corrected gains are 0.188 ± 0.040 and 0.778 ± 0.157 . Their scores are 0.911 and 0.907, with support-aware classes comparison-invalid and comparison-invalid. Accordingly, the new spoke does not supply an established hold/break transition. When a new cell is comparison-invalid, we still report its measured gain but do not convert the discrepancy into a claim that Vlasov physics failed. Their parasitic fractions are 0.268 and 0.189, both above 0.05.

The separately frozen refinement at 0.60 has corrected gain 0.473 ± 0.110 , score 0.912, and class comparison-invalid with illustrative support. It is included in the complete map and compactness audit but is not retroactively inserted into the registered three-point nonlinear-ion conjunction. Its parasitic fraction, 0.211, fails the same prerequisite.

The post-registration boundary-field proxy is 0.42, 0.42, 0.42, and 0.42 at increasing intensity. Because it has no frozen acceptance threshold and is not uniquely electrostatic, it does not alter the map; it instead makes every physical attribution conditional on a future controlled sheath measurement.

As a pipeline diagnostic at the matched 10-MG center, the two cold-basis branch gains are 0.365 ± 0.096 and 0.368 ± 0.100 ; their summed-energy gain is 0.367 ± 0.098 . The largest numerical

Table 3: Conjunctive mechanism tests. Failure of a comparison prerequisite takes precedence over a physical attribution.

segment	hypothesis	status	residual	diagnostic	outcome
near perpendicular ($\phi_B = -13^\circ$)	single IAW/orbits	pre-stated	3/6 break	shape residual 0.78–1.87	3 attributed
high- B , near parallel (50 MG, 75°)	drive window	pre-stated	2/2 break	$\omega_{ce}/\bar{\omega} = 0.16$	2 attributed
intensity spoke	nonlinear ion	pre-stated	2/2 endpoint breaks	$P_{2\Delta k}/P_{\Delta k} = 0.048$ at high I	0 attributed

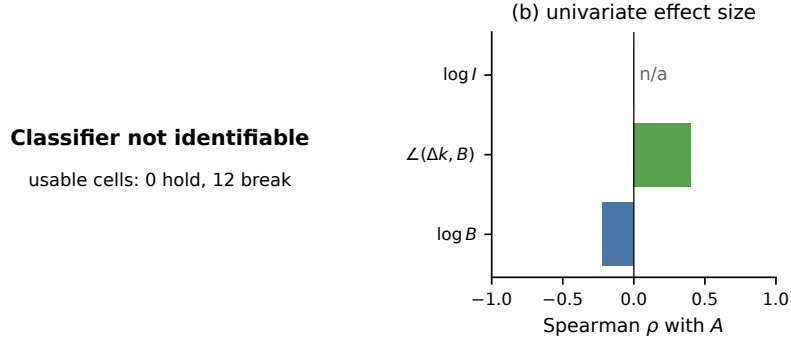


Figure 4: Preregistered compactness audit. The left panel reports classifier identifiability and class counts. Rank correlations are descriptive.

mode-sum energy-closure error is 3.14×10^{-16} . These values verify the filter/shift/separation book-keeping, but are not substituted for the total- field score and are not presented as warm-eigenmode measurements.

4.2 Mechanism attribution

Only 5 cells meet a preregistered physical conjunction. 7 robust breaks are mechanism-unattributed, and 3 cells stop at comparison validity. This low attribution yield is not repaired with a retrospective story. The near-perpendicular segment asks whether line-shape residual accompanies a break; the near-parallel 50-MG segment tests the upper drive-frequency window; and the intensity segment additionally requires the independent ion-density harmonic. Table 3 reports those tests and their pre-stated status.

Specifically, 3 near-perpendicular breaks meet the single-IAW/orbit conjunction and 2 high-field, near-parallel breaks meet the drive-window conjunction. The intensity test attributes zero cells: its registered scores 0.911, 0.901, and 0.907 are not monotone, while the density harmonic ratio decreases from 0.163 at center to 0.048 at high intensity, opposite the pre-stated pattern. The comparison failure already takes precedence, so this failed conjunction is a diagnostic rather than an alternative post-hoc label.

4.3 A compact or entangled boundary?

The frozen Pareto classifier is non-identifiable because its usable set contains 0 holds and 12 breaks. The univariate score correlations are $\rho_B = -0.22$, $\rho_\theta = 0.40$, and $\rho_I = \text{n/a}$. Figure 4 reports that non-identifiability rather than fitting one-class logistic models. Because repeated polarization rows share the three physical coordinates and the sample is small, this is a compression of the mapped domain rather than a universal scaling law.

5 Failure analysis and limitations

First, theoretical provenance is the strongest limitation. We tried publisher retrieval and an independent reconstruction, but could not inspect the companion article’s equations. The resulting solver is magnetized, kinetic, finite-box-aware, and absolutely normalized, yet it fails one published qualitative signature and can have a large absolute bias. Our boundary statements therefore apply to this reproducible reconstruction. A future run of the authors’ exact implementation can reuse the frozen grid and score without reusing these labels for model selection.

Second, the archive gives processed spreads rather than independent PIC replicates. We retain those ranges and add a conservative common-background uncertainty. The fixed-seed processed estimator is exactly reproducible at fixed rank count, as verified by the duplicate, so repeated identical runs would not estimate RNG variance. Full SDF-container hashes differ because file-level metadata are not deterministic; the exact-duplicate claim is only for the processed boundary observable. The rule to switch to clock-seeded triplicates was available only if that observable differed. Thus processed numerical reproducibility is strong but ensemble uncertainty is not measured.

Third, reduced PIC is a validation problem, not merely a cheaper grid. Shorter interaction length changes mode spacing and signal; coarser cells and fewer particles change numerical dispersion and noise. We therefore enforce trend, amplitude, SNR, and 16/16 gates jointly. Any failed drift, depletion, scattering, or convergence test is shown as comparison-invalid rather than silently dropped. The pump-off control is at 10 MG, so its common correction does not identify a possible field-dependent propagation offset. The boundary-normal field ratio partially exposes, but cannot isolate, the unregistered sheath prerequisite; it is therefore not used to rescue or reject a cell after outcomes are known. In fact, the reduced suite cleanly separates numerical validation from theory validity: trend, amplitude, SNR, duplicate, and resolution decisions pass, but the three new intensity conditions have parasitic fractions 0.189–0.268. Their gray cells are therefore an established boundary of the *comparison protocol*, not evidence that a Vlasov term fails.

Fourth, the map is a stated subspace: fixed carbon density, temperatures, crossing angle, and positive 0.9 Å detuning. Field, orientation, and intensity are each sampled at three levels, but the study cannot establish density or temperature boundaries. The archive curves diagnose shape, while new PIC samples only one detuning. Finally, mechanism tests are intentionally conjunctive; 7 unattributed cells show that a residual alone does not identify which term failed first.

6 Computational cost and coverage

The uncalibrated solver evaluates 21 cells, 1278 curve points, and nine dispersion roots in 7.51 s. The full new suite consumes 98.1 core-hours and 14.8 h of concurrent wall time, retaining 1.06 GiB of raw SDF output. Individual measured costs are in Table 4. Five additional base-resolution points would cost approximately 42.9 core-hours at the observed mean of 4.29 h per two-rank run. That budget would most profitably add two intensity-bracketing points and three pump-off or resolution controls; adding five arbitrary core cells would broaden appearance without resolving the present comparison-validity bottleneck.

Table 4: Measured computational cost on the shared host. Concurrent EPOCH wall time was 14.75 h; summed cost was 98.1 core-h.

component	coverage	measured cost
partial Vlasov baseline	21 cells + 1278 curve points	7.51 s wall
EPOCH validation_B2	2 MPI ranks	4.46 h wall
EPOCH validation_B10	2 MPI ranks	4.36 h wall
EPOCH validation_B50	2 MPI ranks	4.39 h wall
EPOCH intensity_I0p25	2 MPI ranks	4.22 h wall
EPOCH intensity_I0p60	2 MPI ranks	3.98 h wall
EPOCH intensity_I1p0	2 MPI ranks	4.28 h wall
EPOCH duplicate_B10	2 MPI ranks	4.25 h wall
EPOCH resolution_B10	2 MPI ranks	14.75 h wall
EPOCH control_pumpoff	2 MPI ranks	4.38 h wall

7 Implications for ICF practice

The map supports a conservative workflow. A Vlasov estimate should first be qualified by the same frequency ratios, beat-wave angle, and intensity regime, then checked against the support-aware color in Fig. 1. Green cells permit this reconstructed baseline at the registered 30% level; red established cells require PIC or a better kinetic implementation; amber cells need reduced uncertainty; gray cells require a better simulation before either model can be judged. This ordering prevents an under-resolved PIC run from being used as evidence against theory.

For the documented reconstruction, the present sample contains no green cell: 12 established red cells require a better kinetic implementation, 6 amber cells require tighter PIC uncertainty, and the 3 new gray cells require suppression or attribution of parasitic light. The map therefore licenses no sampled regime for quantitative ICF use at the 30% level; it is a triage result, not a validated closure.

The practical message is not that magnetization alone sets trust. Orientation controls which ion resonances and orbit assumptions are exposed, polarization controls electromagnetic access to the channel, and intensity controls whether linear comparison remains meaningful. This is consistent with experimental evidence that polarization arrangement changes multibeam CBET [Hao et al., 2025], with Langdon-effect evidence that non-Maxwellian electrons shift ordinary CBET predictions [Turnbull et al., 2020], and with nonlinear simulations that place harmonics and trapping at high intensity [Xu et al., 2026]. The boundary should therefore be used as a triage map, not extrapolated as a universal ICF closure.

8 Conclusion

We have converted a qualitative Vlasov/PIC comparison into a preregistered, uncertainty-aware boundary decision across 21 cells, including raw new EPOCH evidence. The procedure separates physical breaks from invalid comparisons, records failed mechanism conjunctions, and tests low-dimensional compressibility without choosing axes after seeing labels. Its most important negative result is equally explicit: there is no established hold cell for the partial reconstruction, and the new intensity points are comparison-invalid rather than a supported physical boundary. The frozen grid, raw SDF files, and automated analysis make the next comparison decisive: replace the documented reconstruction or extend the validated PIC controls, then rerun the same rule without moving its goalposts.

Reproducibility statement

The preregistration, refinement declaration, source, generated decks, executable hashes, raw SDF files, processed JSON/CSV, figures, LaTeX tables, build log, codebase inspection, and failed-attempt log are all under proposal/. Running `epoch_analysis.py`, `sweep_evaluation.py`, `make_figures.py`, and `make_latex_results.py` regenerates every quantitative paper artifact from the retained inputs. No external nontrivial Vlasov or boundary-analysis code is used.

References

- T. D. Arber, K. Bennett, C. S. Brady, A. Lawrence-Douglas, M. G. Ramsay, N. J. Sircombe, P. Gillies, R. G. Evans, H. Schmitz, A. R. Bell, and C. P. Ridgers. Contemporary particle-in-cell approach to laser-plasma modelling. *Plasma Physics and Controlled Fusion*, 57:113001, 2015. doi: 10.1088/0741-3335/57/11/113001.
- Burton D. Fried and Samuel D. Conte. *The Plasma Dispersion Function*. Academic Press, 1961.
- A. M. Hansen, D. Turnbull, P. Michel, et al. Cross-beam energy transfer saturation by ion heating. *Physical Review Letters*, 126:075002, 2021. doi: 10.1103/PhysRevLett.126.075002.
- A. M. Hansen, K. L. Nguyen, D. Turnbull, B. J. Albright, R. K. Follett, R. Huff, J. Katz, D. Mastroiome, A. L. Milder, L. Yin, J. P. Palastro, and D. H. Froula. Cross-beam energy transfer saturation: ion heating and pump depletion. *Plasma Physics and Controlled Fusion*, 64:034003, 2022. doi: 10.1088/1361-6587/ac493b.

- Liang Hao, Tao Gong, Wen Yi Huo, et al. Revealing crossed-beam energy transfer as the origin of asymmetric backscattering in hohlraums. *Nature Communications*, 16:10343, 2025. doi: 10.1038/s41467-025-65290-1.
- J. A. Marozas, M. Hohenberger, M. J. Rosenberg, D. Turnbull, et al. First observation of cross-beam energy transfer mitigation for direct-drive inertial confinement fusion implosions using wavelength detuning at the national ignition facility. *Physical Review Letters*, 120:085001, 2018. doi: 10.1103/PhysRevLett.120.085001.
- P. Michel, L. Divol, E. L. Dewald, et al. Tuning the implosion symmetry of icf targets via controlled crossed-beam energy transfer. *Physical Review Letters*, 102:025004, 2009. doi: 10.1103/PhysRevLett.102.025004.
- J. D. Moody, P. Michel, L. Divol, R. Berger, et al. Multistep redirection by cross-beam power transfer of ultrahigh-power lasers in a plasma. *Nature Physics*, 8:344–349, 2012. doi: 10.1038/nphys2239.
- J. D. Moody, B. B. Pollock, H. Sio, et al. Increased ion temperature and neutron yield observed in magnetized indirectly driven d2-filled capsule implosions on the national ignition facility. *Physical Review Letters*, 129:195002, 2022. doi: 10.1103/PhysRevLett.129.195002.
- John D. Moody and Yuan Shi. Vlasov theory of magnetized cross-beam energy transfer in a high-energy-density plasma. *Physics of Plasmas*, 33:062712, 2026. doi: 10.1063/5.0322470.
- A. G. Seaton, L. Yin, R. K. Follett, B. J. Albright, and A. Le. Cross-beam energy transfer in direct-drive icf. i. nonlinear and kinetic effects. *Physics of Plasmas*, 29:042706, 2022. doi: 10.1063/5.0078800.
- Yuan Shi. Three-wave interactions in magnetized warm-fluid plasmas: General theory with evaluable coupling coefficient. *Physical Review E*, 99:063212, 2019. doi: 10.1103/PhysRevE.99.063212.
- Yuan Shi. Benchmarking magnetised three-wave coupling for laser backscattering: analytic solutions and kinetic simulations. *Journal of Plasma Physics*, 89:905890305, 2023. doi: 10.1017/S0022377823000405.
- Yuan Shi. Particle-in-cell simulations of magnetized cross-beam energy transfer. Zenodo data set, 2025. doi:10.5281/zenodo.16498564.
- Yuan Shi and John D. Moody. Particle-in-cell simulations of laser crossbeam energy transfer via magnetized ion-acoustic wave. *Physics*, 8(1):25, 2026. doi: 10.3390/physics8010025.
- Yuan Shi, Hong Qin, and Nathaniel J. Fisch. Three-wave scattering in magnetized plasmas: From cold fluid to quantized lagrangian. *Physical Review E*, 96:023204, 2017. doi: 10.1103/PhysRevE.96.023204.
- Thomas H. Stix. *Waves in Plasmas*. American Institute of Physics, 1992.
- David Turnbull, Arnaud Colaitis, Aaron M. Hansen, Avram L. Milder, John P. Palastro, Joseph Katz, Christophe Dorrer, Brian E. Kruschwitz, David J. Strozzi, and Dustin H. Froula. Impact of the langdon effect on crossed-beam energy transfer. *Nature Physics*, 16:181–185, 2020. doi: 10.1038/s41567-019-0725-z.
- Yilin Xu, Yao Zhao, Hongwei Yin, Zhuwen Lin, Yan Yin, Liang Hao, Yaozhi Yi, Hongyu Zhou, Jinlong Jiao, and Anle Lei. Effects of laser-plasma parameters on sub-nanosecond evolution of cross-beam energy transfer. *arXiv preprint arXiv:2508.03152v2*, 2026. doi: 10.48550/arXiv.2508.03152.
- L. Yin, T. B. Nguyen, G. Chen, L. Chacón, D. J. Stark, L. Green, and B. M. Haines. Time-dependent saturation and physics-based nonlinear model of cross-beam energy transfer. *Physics of Plasmas*, 30:042706, 2023. doi: 10.1063/5.0134867.